L.J. INSTITUTES OF COMPUTER APPLICATIONS									
Programme	Master of Computer Applications					**Branch/Spec.**	Computer Applications		
Semester	II					**Effective from Academic Year**	2025-26		
Subject Category	Major					**Effective for the batch Admitted in**	June 2025		
Subject Code	240110201		**Subject Name**			Data Structure (DS)			
Teaching scheme						**Examination scheme (Marks)**			
(Lecture hours Per week)	**Credit**	**L**	**T**	**P**	**Total**	**E**	**CEC**	**V**	**Total**
3	3	3	0	0	36	50	50	0	100

University Exam Evaluation Scheme

Internal Evaluation		**External Evaluation**	
Mid Sem Theory Exam(30 Marks)	**30**	**Theory Exam**	**50**
Attendance	**10**	**Practical Exam**	**NA**
Theory Assignment	**10**		
Total Marks	**50**	**Total Marks**	**50**

Pre-requisites:

- Basic knowledge of writing and understanding algorithms for solving a problem.
- Knowledge of programming in C language

Program Learning Outcome:

Name of PO	**Description**
PO1	Foundational & Theoretical Mastery
PO2	Software Design & Development Expertise
PO3	Problem-Solving & Analytical Ability
PO4	Proficiency in Emerging Technologies
PO5	Information Systems & Management Perspective
PO6	Professional & Ethical Responsibility
PO7	Communication & Teamwork
PO8	Lifelong Learning & Career Readiness

Course Learning Outcome:

Name of CO	**Description**
CO1	To understand how to solve a problem step by step to get the desired output with different way.

CO2	To learn the optimal way to organize data in the digital space.
CO3	To understand the fundamental concept of elementary data structures and their implementation.
CO4	To understand where and how the data structures are implemented to solve the real world problems.
CO5	To learn how to write efficient and optimized computer programs.

Mapping of CO and PO:

Course Learning Objectives	Program Outcomes (POs)							
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	2	2	-	-	-	-	1
CO2	2	3	3	-	-	-	-	1
CO3	2	3	3	-	2	2	-	1
CO4	2	3	3	2	-	2	-	1
CO5	1	3	2	2	-	-	2	1

Content:

Unit	Content	Weightage (%)	No of Hrs.	CO
Unit I	**Introduction to Data Structures** Data types – primitive and non-primitive, abstract data type, concept of data structures, what is algorithm?, introduction to linear and nonlinear data structures, introduction to calculation of time complexity (best case, worst case and average case) and space complexity.	**15%**	**4**	**CO1**
Unit II	**Linear Data Structures: Array, Linked lists, Stacks, Queues** **Array:** Operations (row major and column major address calculations) **Linked List: B**asic concept, difference between linked list and array, storage representation, types of linked lists (singly, doubly, circular), operations on linked list (insert, modify, delete, union, intersection, merge, sort, searching, reverse (by changing address not data)), applications of linked list (polynomial operations – addition), sparse matrices (only concept).	**20%**	**6**	**CO2**
Unit III	**Stack:** Basic concept, storage representation (array and linked list), basic operations (push, pop, peep and change), applications of stacks (polish and reverse polish expressions), arithmetic	**20%**	**4**	**CO3**

	expression evaluation using stack. **Queues**: Basic concept, storage representation (array and linked list), basic operations (insert and delete). Types of queues- circular, deque, priority queues (only concept). Application of queues.			
Unit IV	**Non Linear Data Structures – Trees and Graphs.** **Trees** – Basic concept, terms associated with trees (Node, parent, child, link, root, leaf, level, height, indegree, outdegree, siblings), Storage representation – Linear and Linked, Conversion of General tree to Binary tree, Complete Binary tree, full binary tree, BST operations, Tree traversals – In-order, Pre-order, Post-order, Types of Binary tree : BST (Practical – recursive), Expression tree, AVL Tree (without practical),2-3 tree, B tree(without practical)) **Graphs**-Basic concepts, technical terms associated with Graphs – Digraph, Weighted graph, adjacent vertices, self loop, parallel edges, simple graph, complete graph, isolated vertex, Degree of a vertex, connected graph-, Storage representation (Set representation, Adjacency matrix, Adjacency list), Graph Traversing algorithms- DFS and BFS. Spanning Tree, Minimum spanning tree algorithm: Prim's Algorithm, Kruskal's Algorithm, Shortest path algorithm – Dijikstra 's Algorithm (without practical)	**30%**	**6**	**CO3**
Unit V	**Sorting and Searching Algorithms and Hashing** **Sorting Algorithms:** Bubble, Selection, Insertion, Shell, 2- Way Merge sort, Radix sort, heap sort and Quick sort. **Searching Algorithms:** Binary Search, **Hashing:** Introduction to Hashing and hashing methods, collision resolution techniques.	**15%**	**4**	**CO4**

Reference Books:	
1	Debasis Samanta, Classic Data Structures , PHI, Second Edition.
2	Jean-Paul Tremblay, Paul G. Sorenson, "An Introduction to Data Structures with Applications", Tata McGraw-Hill, 2nd Edition, (2007).
3	Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", Pearson, Second Edition,
4	Ashok N. Kamthane, "Introduction to Data Structures in C", Pearson Education (2004).
5	Reema Thareja, Data Structures using C, Oxford
Web Reference:	
1	**https://www.tutorialspoint.com/data_structures_algorithms**
2	**https://www.javatpoint.com/**
3	**GeeksforGeeks (Data Structures & Algorithms):** https://www.geeksforgeeks.org

Question Paper Scheme:	
	University Examination Duration: 2 Hours 30 minutes. Q-1: 10 Marks Q-2: 10 Marks Q-3: 10 Marks Q-4: 10 Marks Q-5: 10 Marks